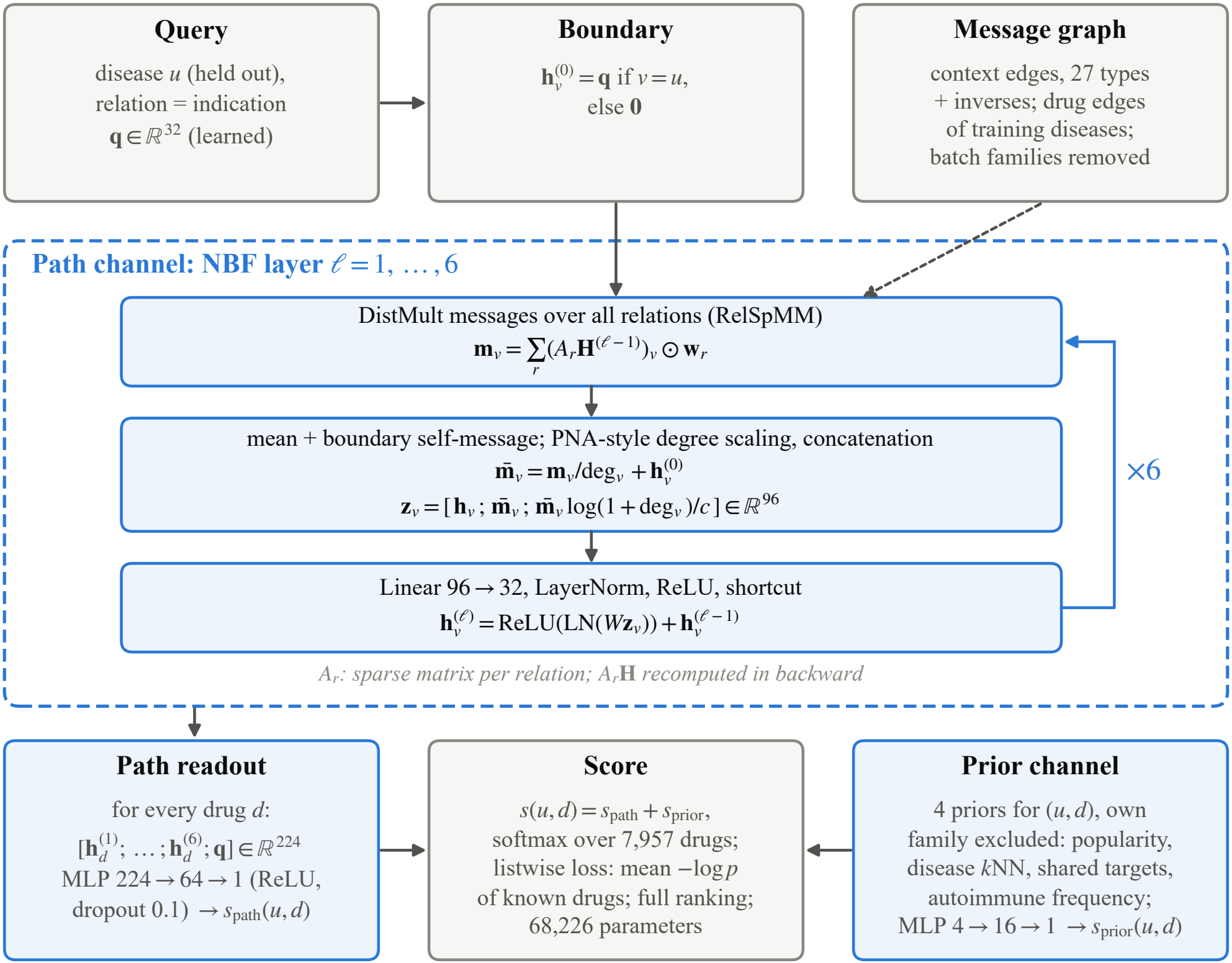

Query
disease u (held out),
relation = indication
$\mathbf{q} \in \mathbb{R}^{32}$ (learned)
Boundary
$\mathbf{h}_v^{(0)} = \mathbf{q}$ if $v = u$,
else $\mathbf{0}$
Message graph
context edges, 27 types
+ inverses; drug edges
of training diseases;
batch families removed
Path channel: NBF layer $\ell = 1, \ldots, 6$
DistMult messages over all relations (RelSpMM)
$\mathbf{m}_v = \sum_r (A_r \mathbf{H}^{(\ell-1)})_v \odot \mathbf{w}_r$
mean + boundary self-message; PNA-style degree scaling, concatenation
$\bar{\mathbf{m}}_v = \mathbf{m}_v / \deg_v + \mathbf{h}_v^{(0)}$
$\mathbf{z}_v = [\mathbf{h}_v ; \bar{\mathbf{m}}_v ; \bar{\mathbf{m}}_v \log(1+\deg_v)/c] \in \mathbb{R}^{96}$
Linear $96 \to 32$, LayerNorm, ReLU, shortcut
$\mathbf{h}_v^{(\ell)} = \mathrm{ReLU}(\mathrm{LN}(W\mathbf{z}_v)) + \mathbf{h}_v^{(\ell-1)}$
×6
A_r: sparse matrix per relation; $A_r\mathbf{H}$ recomputed in backward
Path readout
for every drug d:
$[\mathbf{h}_d^{(1)}; \ldots; \mathbf{h}_d^{(6)}; \mathbf{q}] \in \mathbb{R}^{224}$
MLP $224 \to 64 \to 1$ (ReLU,
dropout 0.1) $\to s_{\text{path}}(u, d)$
Score
$s(u, d) = s_{\text{path}} + s_{\text{prior}}$,
softmax over 7,957 drugs;
listwise loss: mean $-\log p$
of known drugs; full ranking;
68,226 parameters
Prior channel
4 priors for (u, d), own
family excluded: popularity,
disease kNN, shared targets,
autoimmune frequency;
MLP $4 \to 16 \to 1$ $\to s_{\text{prior}}(u, d)$